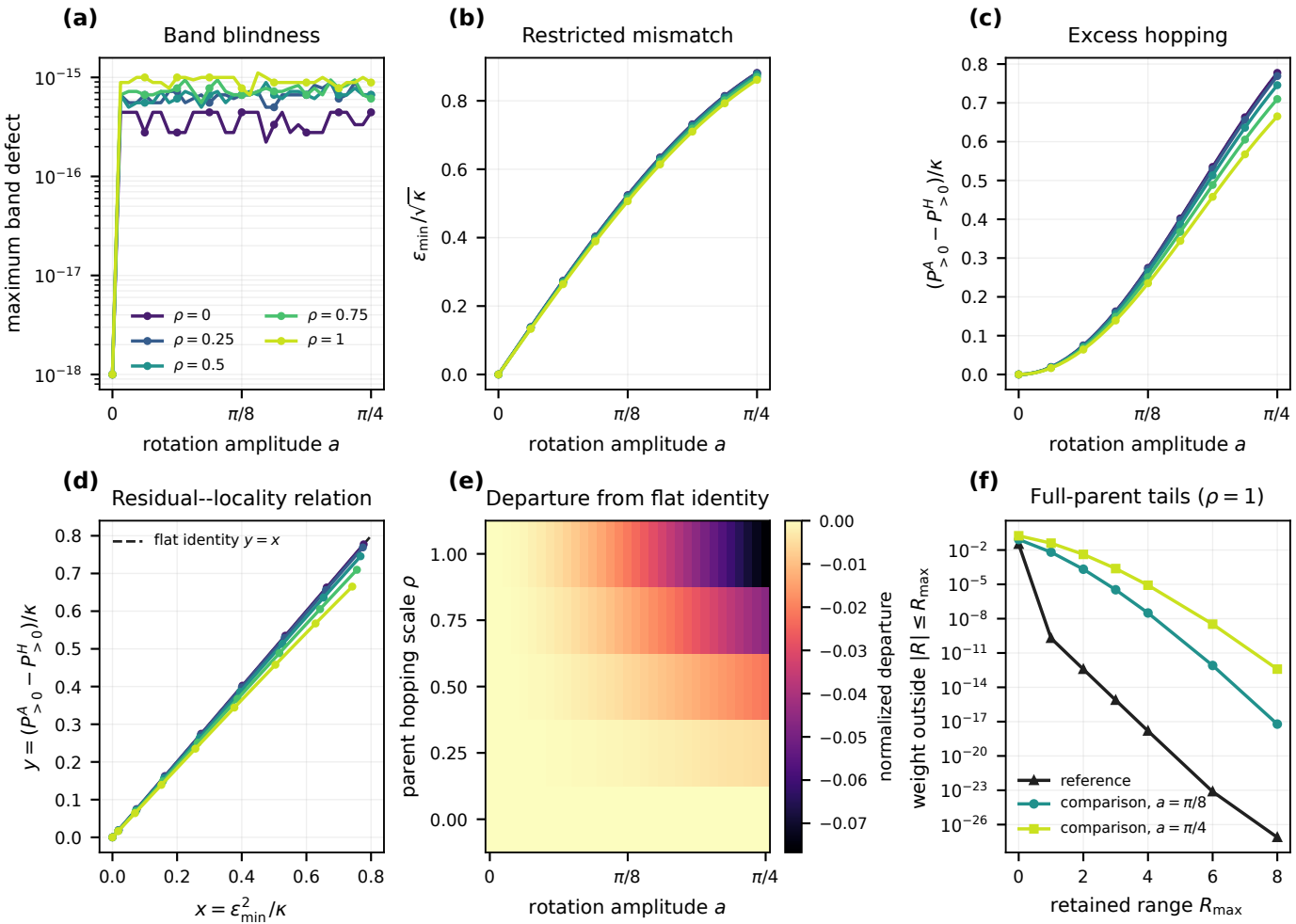

(a)
Band blindness
maximum band defect
10^{-15}
10^{-16}
10^{-17}
10^{-18}
$\rho = 0$
$\rho = 0.25$
$\rho = 0.5$
$\rho = 0.75$
$\rho = 1$
0
π/8
π/4
rotation amplitude a
(b)
Restricted mismatch
$\varepsilon_{\min}/\sqrt{\kappa}$
0.0
0.2
0.4
0.6
0.8
0
π/8
π/4
rotation amplitude a
(c)
Excess hopping
$(P^A_{>0} - P^H_{>0})/\kappa$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
π/8
π/4
rotation amplitude a
(d)
Residual--locality relation
flat identity $y = x$
$y = (P^A_{>0} - P^H_{>0})/\kappa$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
$x = \varepsilon^2_{\min}/\kappa$
(e)
Departure from flat identity
parent hopping scale ρ
0.00
0.25
0.50
0.75
1.00
0
π/8
π/4
rotation amplitude a
normalized departure
0.00
−0.01
−0.02
−0.03
−0.04
−0.05
−0.06
−0.07
(f)
Full-parent tails ($\rho = 1$)
weight outside $|R| \le R_{\max}$
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-14}
10^{-17}
10^{-20}
10^{-23}
10^{-26}
reference
comparison, $a = \pi/8$
comparison, $a = \pi/4$
0
2
4
6
8
retained range $R_{\max}$